WHITE PAPER · AI IN ASEPTIC DRUG PROCESSING · CONTAMINATION CONTROL

Beyond the Threshold: *The Shift to Aseptic Predictive Intelligence*

Aseptic Predictive Intelligence uses machine learning and Bayesian mathematics to classify and locate contamination sources in real-time — preventing batch rejections before a threshold is ever breached.

THE CHALLENGE

The "Silent Breach" Problem

Threshold Monitoring is Reactive

Traditional systems only trigger after limits are breached, leading to manual, "guesswork" investigations.

Data-Rich but Insight-Poor

Thousands of particle counts are logged every hour, yet their specific causes remain unidentified.

Contamination as a Signal

Contamination is not random — it is a "*Digital Fingerprint*" moving through a *fluid-dynamic grid*. Every particle has a name and a source.

Case Study — Invisible Bearing Failure (ISO 5 Filling Line)

85% Faster Response

Hybrid Forensic Model vs. Traditional SPC (Threshold-Based)

Traditional SPC — **14.5 min — Batch at risk**

Predictive Intelligence — **2.2 min — Batch preserved**

THE SOLUTION

The Dual-Engine Framework

⏱ **Temporal Engine — WHAT HAPPENED?**

Uses sktime to identify "Shapelet Signatures"

Unique discriminative patterns in time series data that identify the contamination mechanism — whether glove breach, mechanical friction, or HVAC drift — while the event is still unfolding.

Glove Breach

Sharp rise, slow decay

Instantaneous release followed by HVAC dilution

Mechanical Friction

Rhythmic sawtooth

Periodic emissions tied to component RPM/cycles

HVAC Drift

Gradual baseline elevation

Slow, uniform increase indicating filter degradation

↗ **Spatial Engine** — WHERE IS IT?

Bayesian Search Theory — Source Triangulation

Sensor data and airflow dynamics are fed into a Bayesian probability grid. "Negative evidence" (clean upstream sensors) prunes impossible locations, narrowing a 3-metre filling line to a **20 cm target zone** — instantaneously.

S_1 — 2%

S_2 — 3%

S_3 — 11%

S_4 — 78%★

S_5 — 5%

S_6 — 1%

★ Posterior probability converges at Sensor 4. Upstream sensors (S_1–S_3) provide negative evidence, eliminating upstream source hypotheses. No operator calculation required.

"Every particle has a name and a source. The data to find both is already being collected."